

Complex Analysis Exam Problems

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Fall 2025

Part I

1. Define $\gamma: [0, 2\pi] \rightarrow \mathbb{C}$ by

$$\gamma(t) = e^{it}.$$

Compute

$$\int_{\gamma} z^n dz$$

for every integer n .

Solution. Since

$$\gamma(t) = e^{it}, \quad 0 \leq t \leq 2\pi,$$

we have

$$\gamma'(t) = ie^{it}.$$

Therefore,

$$\int_{\gamma} z^n dz = \int_0^{2\pi} \gamma(t)^n \gamma'(t) dt = i \int_0^{2\pi} e^{i(n+1)t} dt.$$

If $n \neq -1$, then

$$\begin{aligned} \int_{\gamma} z^n dz &= i \left[\frac{e^{i(n+1)t}}{i(n+1)} \right]_0^{2\pi} \\ &= \frac{e^{2\pi i(n+1)} - 1}{n+1}. \end{aligned}$$

Because $n+1$ is an integer,

$$e^{2\pi i(n+1)} = 1,$$

and hence

$$\int_{\gamma} z^n dz = 0.$$

If $n = -1$, then

$$\int_{\gamma} z^{-1} dz = i \int_0^{2\pi} 1 dt = 2\pi i.$$

Thus

$$\int_{\gamma} z^n dz = \begin{cases} 2\pi i, & n = -1, \\ 0, & n \neq -1. \end{cases}$$

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2. Let γ be the closed polygon

$$[1 - i, 1 + i, -1 + i, -1 - i, 1 - i].$$

Find

$$\int_{\gamma} \frac{1}{z} dz.$$

3. 1. Show that a Möbius transformation has 0 and ∞ as its only fixed points if and only if it is a dilation, but not the identity.
2. Show that a Möbius transformation has ∞ as its only fixed point if and only if it is a translation, but not the identity.
4. Let G be a region, and suppose that $f: G \rightarrow \mathbb{C}$ is analytic such that $f(G)$ is a subset of a circle. Show that f is constant.

5. Let

$$f(z) = e^x \sin y + ie^x \cos y, \quad z = x + iy.$$

Determine whether f is analytic on the complex plane.

6. Map

$$G = \mathbb{C} \setminus \{z : -9 \leq z \leq 9\}$$

onto the open unit disk by an analytic function f . Can f be one-to-one?

7. Give the principal branch of

$$\sqrt{1 - z^3}.$$

Part II

8. Compute the integral

$$\int_{\gamma} \frac{e^z}{(z - 2)^2(z - 5)^5} dz,$$

where γ is the closed path

$$\gamma(t) = 3e^{it}, \quad 0 \leq t \leq 2\pi.$$

9. Compute the integral in the previous exercise with γ the closed path

$$\gamma(t) = e^{it}, \quad 0 \leq t \leq 2\pi.$$